

STANDARD OPERATING PROCEDURE

APPLICATION DEPLOYMENT / RELEASE

REVISION HISTORY

Revision No. / Version No.	Date	Prepared by	Approved by	Details of Change
1.0	28-Jul-2022	Anant Yende (CTO)		
2.0	02-Aug-2022	Anant Yende (CTO)		
3.0	28-Feb-2023	Prachi Naik (Asst. Manager IT)	Anant Yende (CTO)	
4.0	20-Jan-2025	Austin De Sa (SSD)	Prachi Naik (Asst. Manager IT)	Added note in Procedure to follow & Responsibilities section

OBJECTIVE

To ensure planned software release.

PARTICIPANTS

1. Deployment Lead:
2. Deployment Engineer: The engineer allotted for deployment. Default TL unless otherwise specified vide office order.
3. Technical Lead (TL): TL who merges the code for deployment.

SCHEDULE OF DEPLOYMENT

Frequency: Every Tuesday & Friday (hereinafter referred as 'D'). In case if it is a holiday than deployment to be done on previous working day. In case of emergency deployment, request needs to be raised to System Architect / Technical Resource Manager by respective PM/TL.

Deployable build time: 12.00 pm to 12.30pm **Deployment time:** 6pm.

PROCEDURE TO FOLLOW:

Release management & freezing of code

1. On (D-1) day (or on the deployment day in case of ad-hoc deployment),
Deployment Lead or Deployment Engineer will create release management entry in
GitLab towards the upcoming deployment.
2. The release entry will be as per mentioned format
<<Project>>_ddMMMyyy e.g. GoaServices_09Apr2021.
3. Once the release entry is created, Deployment Lead will put a message in GEL Technical Group
informing all TLs to give details of their respective project deployment to be included in the
release. All the respective TLs will add the details about the changes/new development they
have committed for production release in the release entry in GitLab.
4. Respective TL to commit code and verify there are no merge errors by 1PM & enter the details in
release entry in GitLab mentioning details in below mentioned format
**<TL Name> - <Service/Project Name> - <Functionality added/changed> -
< [Comma separated List of pages/classes added or modified]>**
e.g.
Prachi - DSP - Change in webservice call for NIC ws - [digitalsurvey.java,
bll_digitalsurvey]
5. Deployment Lead or Deployment Engineer will pull the code from GitLab at 3PM on (D- 1) day
and add the list of files suggested by GITLAB which are modified/added to the release entry in
GitLab.
6. The merge and release entry will be freezed at 3PM on (D-1) day.
7. The Deployment Lead will post the complete release entry from GitLab text message on (D-1)
day in GEL TECHNICAL Group for information/objections by other TLs by 4 PM.
8. In case of any issues pointed by the TLs in the release entry about files being wrongly merged
the same will be reverted by respective TL and inform the Deployment Engineer to take a fresh
pull and update the release entry.

Note: In case the Deployment Lead is on leave on the deployment day, alternate Deployment Lead should be available for deployment. Also, in case of emergency deployment, request needs to be raised to System Architect / Technical Resource Manager by respective PM/TL.

Deployment day

1. Any Deviation in release note published should be confirmed with CTO by the respective TL/PM and if approved get the list of files to be added/removed shared with CTO for onward sharing with Deployment Engineer.
2. Based on the approval the Deployment Engineer will need to take a fresh pull and publish the updated release entry from GitLab in the GEL TECHNICAL GROUP.
3. Deployment Engineer will then copy the build on production server, Keep the production backup ready between 3 PM to 5.30 PM on day D.
4. Code deployment to be done exactly at 6:00 pm.
5. Moving the deployables into root directory for deployment.
6. Message needs to be put in the GEL technical group before & after deployment.
7. After deployment, Deployment Lead / Deployment Engineer will check if site is working or not by logging in.
8. Deployment Lead will then put a message in the GEL Technical Group after the deployment.
9. Respective TL(s) who have merged their code, will have to check if their code changes is working or not.
10. In case issues occur due to deployment; TL will first check if the issue can be resolved without a redeployment.
11. In case a redeployment is needed; TL will fix the issue and merge the code again.
In such cases, deployment will have to be done by the TL itself.
12. If the issue cannot be fixed, the deployed code will be reverted to the previous version and Deployment Lead will inform on GEL Technical Group about failed deployment.

RESPONSIBILITIES

1. Deployment Planning

- a) Creation of release management entry in GitLab.
- b) Putting message in GEL Technical Group that the release has been created.

2. Build and Release Management:

- a) Pulling the code and checking for any issues in the code.
- b) Reporting issues if any to resolve or remove the affected code from the release.

3. Documentation:

- a) Ensuring change log has been added to the final release.
- b) Sharing the final release document on the GEL Technical Group.

4. Deployment Preparation:

- a) Copying the build on production server, and keeping the production backup ready.
- b) Moving the build from building environment to live environment for deployment.
- c) Ensuring backup is taken of existing deployed code on live environment.

5. Deployment Process:

- a) Informing about downtime of application during deployment on the GEL Technical Group.
- b) Taking backup of existing properties / configuration file to be replaced in deployed code.
- c) Moving the deployables into root directory for deployment.
- d) Replacing the properties / configuration files into deployed build code.

6. Testing:

- a) After deployment testing if site is working or not by logging in.
- b) On failure, use the backup build code and inform on GEL Technical Group about failed deployment.
- c) Informing teams regarding the deployment.

7. Carry out deployment / connect to server for deployment only from system with antivirus latest updates.

8. Maintain secrecy of the credentials of the server.

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APPLICATION DEPLOYMENT / RELEASE

LOGGING IN JIRA/TM

Issue Type*: Deployment

Assignee*: Deployment Engineer

Reporter (ticket to be created by) *: Deployment Engineer

Description: Release note details.

Priority*: Medium

Watcher: CTO, Deployment Lead

Original Estimate*: 30 min

Due Date*: Same day

Work Logging by*: Deployment Engineer

Status: Close after deployment-by-Deployment Engineer

Project: In case of multiple then GO/GS main project otherwise respective projects.